

Let:

$$\vec{b}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 5 \\ -6 \end{bmatrix}, [\vec{x}]_S = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, B = \{\vec{b}_1, \vec{b}_2\}$$

$[\vec{x}]_S$ simply means $\vec{x}$ in standard form, we could just have written $\vec{x}$, but for the sake of this example I want to be abundantly clear.

The set B is an alternative basis to $\mathbb{R}^2$.

Theorem

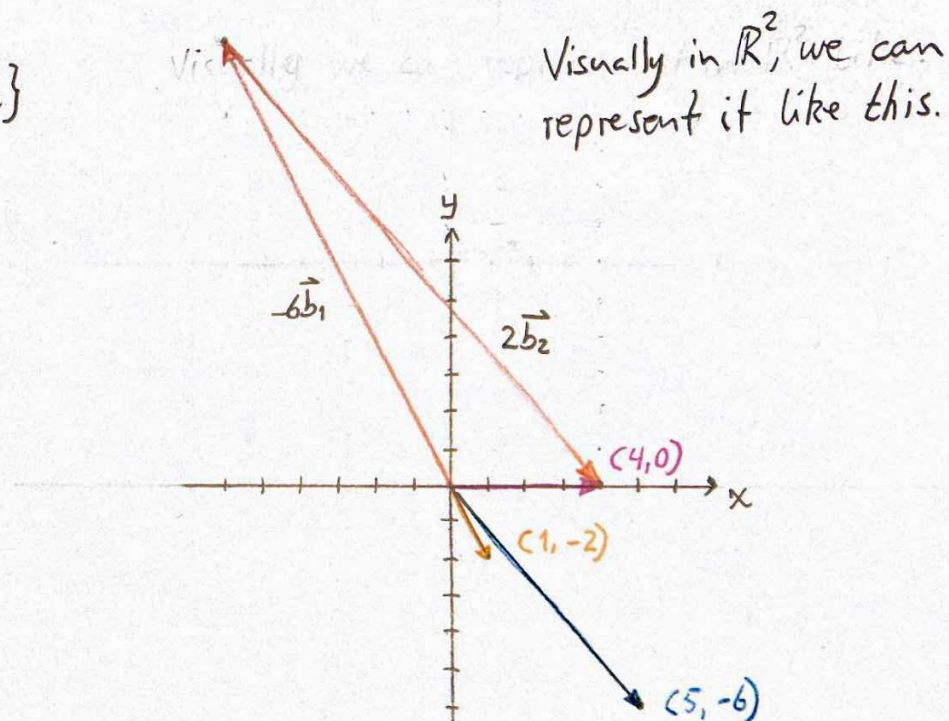
If V is a vector space and

$$B = \{\vec{b}_1, \vec{b}_2, \dots, \vec{b}_n\}$$

is a basis for V , then every $\vec{x} \in V$ can be written uniquely as

$$\vec{x} = c_1 \vec{b}_1 + c_2 \vec{b}_2 + \dots + c_n \vec{b}_n$$

Suppose we want to represent $\vec{x}$ in terms of V 's basis vectors; i.e. how much of $\vec{b}_1$ and $\vec{b}_2$ do we need? This can be found by setting up the system in an augmented matrix and perform reduction.



Setting up the augmented matrix

$$\begin{bmatrix} 1 & 5 & 4 \\ -2 & -6 & 0 \end{bmatrix}$$

After row reduction the rref matrix gives c_1 and c_2 directly.

$$R_2 \leftarrow R_2 + 2R_1$$

$$R_2 \leftarrow R_2 \cdot 1/4$$

$$\begin{bmatrix} 1 & 5 & 4 \\ 0 & 1 & 2 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - 5R_2$$

$$\begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & 2 \end{bmatrix}$$

If we say:

$$[\vec{x}]_S = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

Coefficients (or coordinates) relative to standard basis (origin)

$$[\vec{x}]_B = \begin{bmatrix} -6 \\ 2 \end{bmatrix}$$

Coefficient relative to B

$$\vec{x} = -6\vec{b}_1 + 2\vec{b}_2$$

Actual vector x expressed using basis B .

MAT 3800 Two examples - address system in $\mathbb{R}^2$

II

Let's now look at the same thing in a different way.

From the theorem we have:

$$\vec{x} = c_1 \vec{b}_1 + c_2 \vec{b}_2 + \dots + c_n \vec{b}_n$$

From this we can say:

$$[\vec{x}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Now we can write:

$$[\vec{x}]_S = \underbrace{[\vec{b}_1 \mid \vec{b}_2 \mid \dots \mid \vec{b}_n]}_{P_B} \cdot \underbrace{\begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}}_{[\vec{x}]_B} := P_B [\vec{x}]_B$$

And here it gets interesting:

$$[\vec{x}]_S = P_B [\vec{x}]_B \rightarrow \text{If we now multiply both sides by } P_B^{-1} \text{ we have an equation for } [\vec{x}]_B$$

$$[\vec{x}]_B = P_B^{-1} [\vec{x}]_S$$

Let's first take a moment to understand what P_B is.

P_B is a matrix where the vectors in B are columns.

So for our example we have:

$$P_B = \begin{bmatrix} 1 & 5 \\ -2 & -6 \end{bmatrix}$$

How can we find the inverse? Well, I will give you the 'matrix' way - or Gauss-Jordan elimination:

$$\left[\begin{array}{cc|cc} 1 & 5 & 1 & 0 \\ -2 & -6 & 0 & 1 \end{array} \right]$$

$$R_2 \leftarrow R_2 + 2R_1$$

$$R_2 \leftarrow R_2 \cdot 1/4$$

$$\left[\begin{array}{cc|cc} 1 & 5 & 1 & 0 \\ 0 & 1 & 1/2 & 1/4 \end{array} \right]$$

$$R_1 \leftarrow R_1 - 5R_2$$

$$\left[\begin{array}{cc|cc} 1 & 0 & -3/2 & -5/4 \\ 0 & 1 & 1/2 & 1/4 \end{array} \right] P_B^{-1}$$

$$P_B^{-1} [\vec{x}]_S = \begin{bmatrix} -3/2 & -5/4 \\ 1/2 & 1/4 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -6 \\ 2 \end{bmatrix}$$